

INTRODUCTION

-JAVA PROGRAMMING

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Objectives

- Programming.
- Object Oriented Programming.
- Application of OOP.
- What is Java?
- History of JAVA.
- Why JAVA is called...
- Basic Overview of JAVA.

Programming

- One thing that you will learn quickly is that a computer is very dumb, but obedient.
- It does exactly what you tell it to do, which is not necessarily what you wanted.
- Programming will help you learn the importance of clarity of expression.

“It has often been said that a person does not really understand something until he teaches it to someone else. Actually a person does not really understand something until after teaching it to a computer, i.e., express it as an algorithm.”

81) -(Donald Knuth, in "American Mathematical Monthly,"

What is Computer Programming?

- Computer programming (often shortened to programming) is a process that leads from an original formulation of a computing problem to executable computer programs.
- Programming is the process of taking functional processing steps (algorithm) and encode it into programming language.

Types of programming.

1. Procedure Oriented Programming.
2. Object Oriented Programming.

Object-oriented programming

- What is Object?
- What is Class?
- Apple is a fruit.
- Features of OOP.

Features of OOP

- Encapsulation
- Data Abstraction
- Inheritance
- Polymorphism
- Extensibility
- Persistence
- Delegation
- Genericity
- Object Concurrency
- Event Handling
- Message Passing



Application of OOP

- Image Processing.
- Pattern Recognition.
- Intelligent Systems.
- Database Management System.
- Web based Application.
- Distributed Computing & Applications.
- Enterprise Resource Planning.
- Mobile Computing.
- Data warehousing & Data Mining.
- Parallel Computing

What is JAVA?

- **Java** is a set of computer software and specifications (developed by Sun Microsystems, which was later acquired by the Oracle Corporation) that provides a system for developing application software and deploying it in a cross-platform computing environment.
- Java is a General Purpose, class based, object oriented, Platform independent, portable, Architecturally neutral, multithreaded, dynamic, distributed and robust interpreted Programming Language.
- Write once, Run Anywhere(WORA)

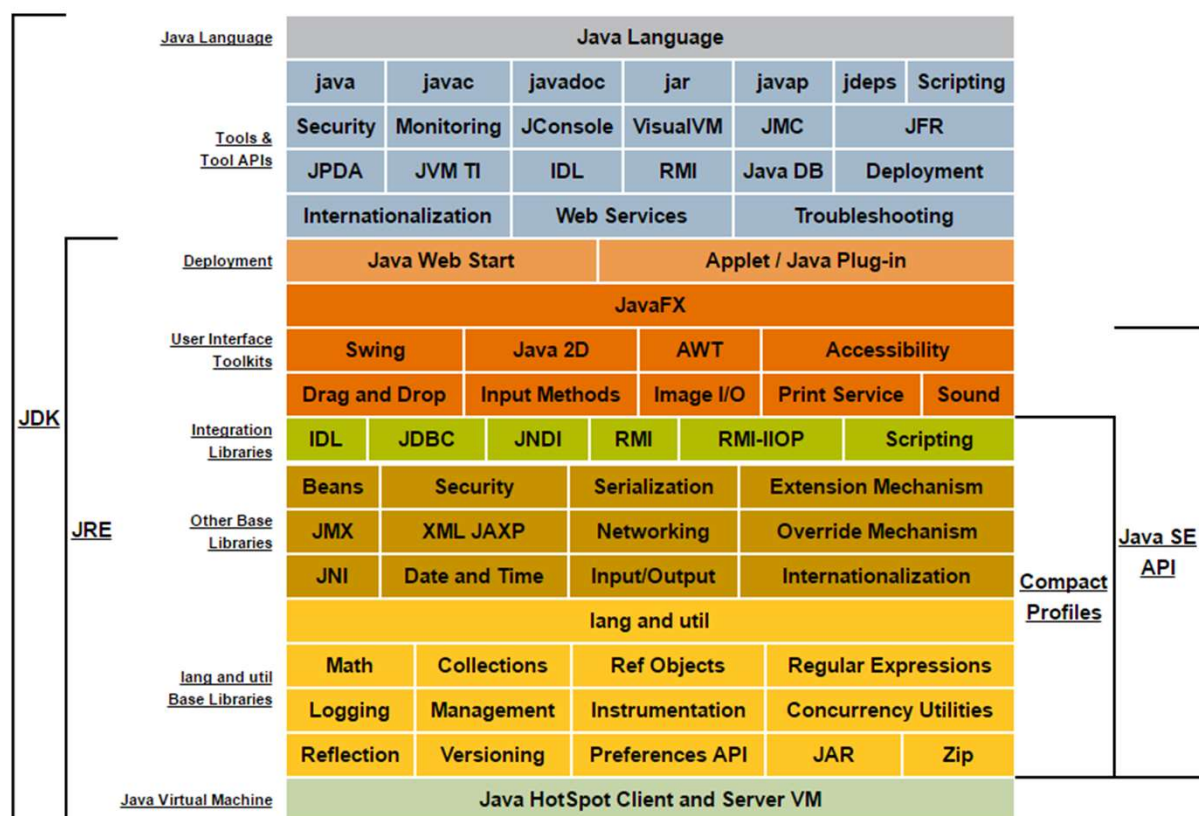
History of JAVA

Version	Release Date	Highlights / Key Features
Java 1.0	Jan 1996	Initial release; AWT, Applets, basic Java APIs.
Java 1.1	Feb 1997	JDBC, RMI, inner classes, JavaBeans, Reflection.
Java 1.2 (J2SE)	Dec 1998	Swing, Collections Framework, security, JIT compiler.
Java 1.3 (J2SE)	May 2000	HotSpot JVM default, JavaSound API, minor performance upgrades.
Java 1.4 (J2SE)	Feb 2002	assert, NIO, Logging API, XML parsing, exception chaining.
Java 5.0 (1.5)	Sept 2004	Generics, Annotations, Enums , Autoboxing, varargs, enhanced for loop.
Java 6 (1.6)	Dec 2006	Scripting support, performance improvements, web services, compiler API.
Java 7	July 2011	NIO.2, try-with-resources, switch on strings, Diamond operator, Fork/Join framework.
Java 8 (LTS)	Mar 2014	Lambda, Streams, Date & Time API , Optional, default methods.
Java 9	Sept 2017	Modules (Project Jigsaw) , JShell (REPL), improved JDK structure.
Java 10	Mar 2018	var keyword (local variable inference), application class-data sharing.
Java 11 (LTS)	Sept 2018	HTTP Client API, var in lambdas, removed Java EE, flight recorder.
Java 12	Mar 2019	Switch expressions (preview), microbenchmarking, Shenandoah GC.
Java 13	Sept 2019	Text blocks (preview), dynamic CDS archives.
Java 14	Mar 2020	Records (preview), helpful NPEs, pattern matching (preview).
Java 15	Sept 2020	Text blocks (standard), sealed classes (preview), hidden classes.
Java 16	Mar 2021	Records (standard), pattern matching (instanceof), vector API (incubator).
Java 17 (LTS)	Sept 2021	Sealed classes (standard), foreign function API (incubator), strong encapsulation.

History of JAVA

Version	Release Date	Highlights / Key Features
Java 18	Mar 2022	Simple web server, UTF-8 as default charset, code snippets in javadoc.
Java 19	Sept 2022	Virtual threads (Project Loom) (preview), structured concurrency (preview).
Java 20	Mar 2023	Record patterns, scoped values, switch pattern matching (preview).
Java 21 (LTS)	Sept 2023	Virtual threads (final) , pattern matching, string templates (preview), sequenced collections.
Java 22	Mar 2024	String templates (2nd preview), unnamed classes (preview), foreign function & memory API (finalized).
Java 23	Sept 2024	Preview of primitive types in generics, stable pattern matching for switch and record patterns, and continued enhancements to virtual threads and foreign function APIs.
Java 24	Mar 2025	Finalized foreign function and memory API, unnamed classes (preview), string templates (second preview), and continued improvements to virtual threads and structured concurrency.
Java 25 (Expected, LTS)	Sept 2025	Next LTS release after Java 21 , likely to include stable string templates, structured concurrency, and improved memory model.

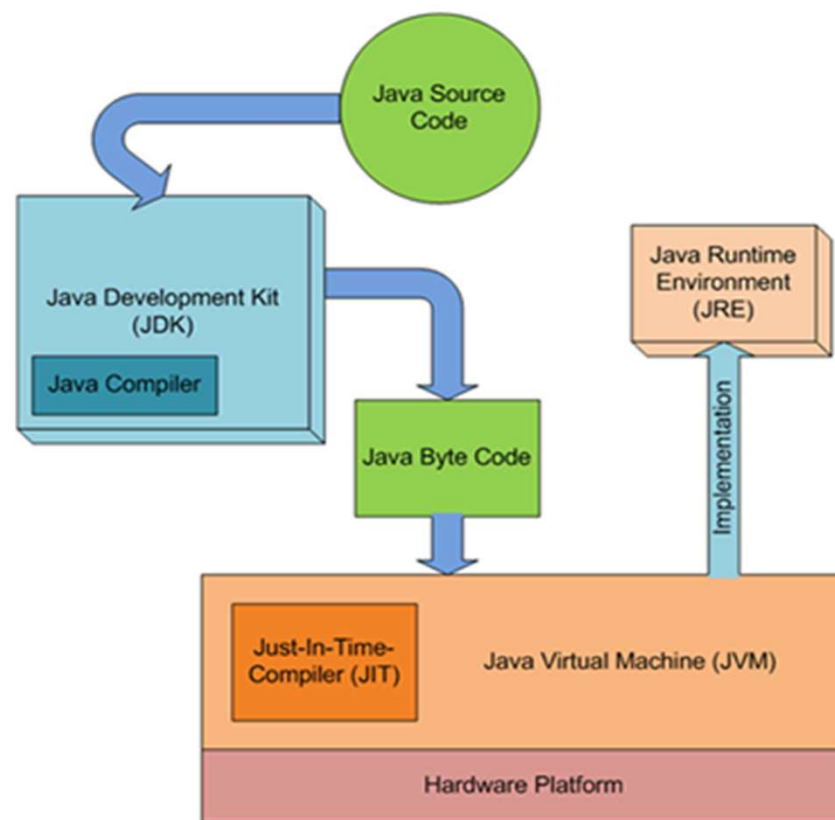
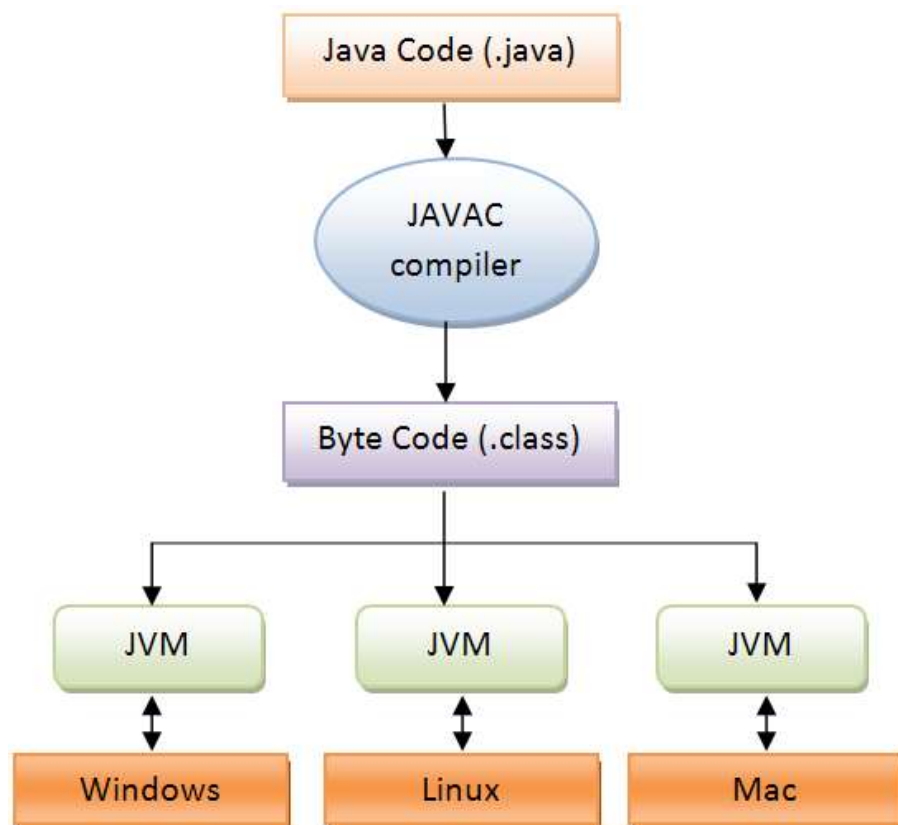
Java Components



Why java is called:

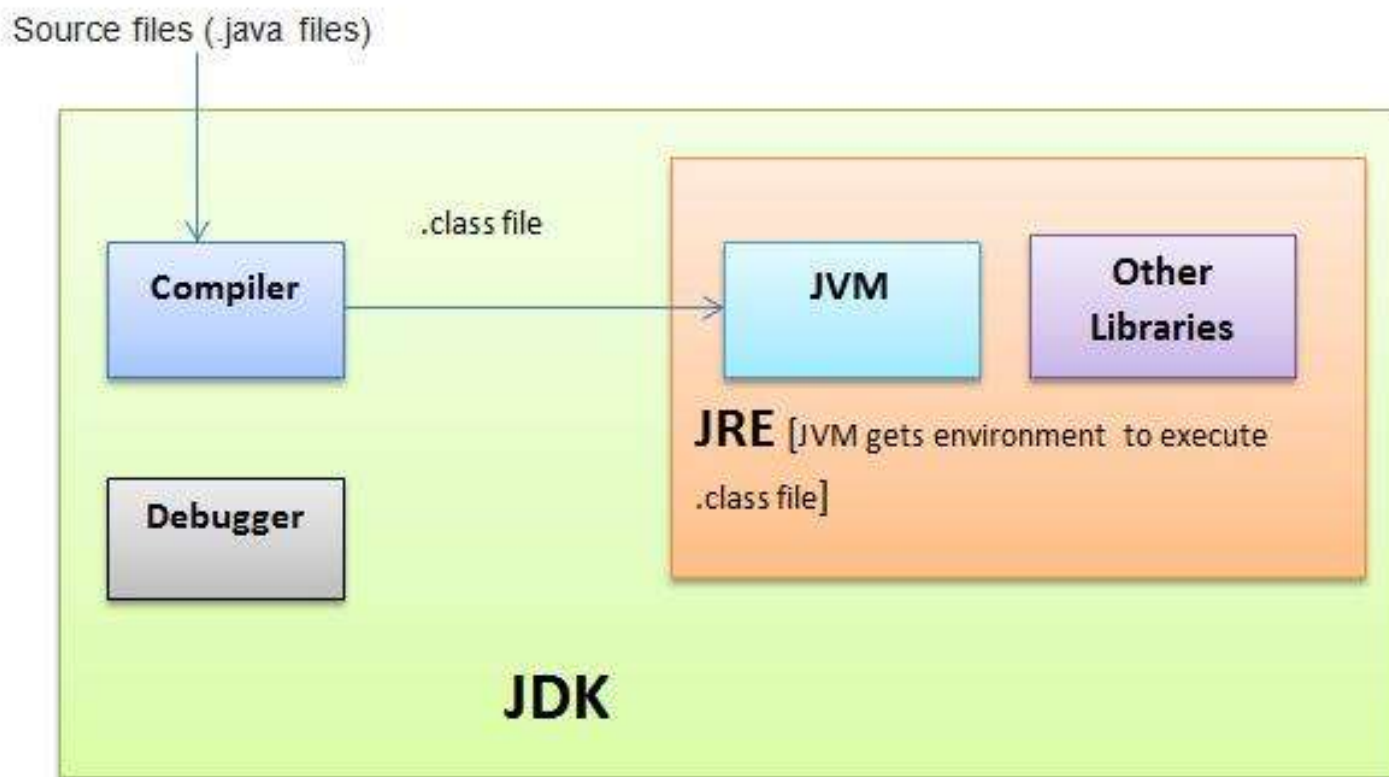
- General Purpose
- Class based
- Object oriented
- Platform Independent
- Architecturally Neutral
- Dynamic
- Distributed
- Portable
- Robust
- Interpreted
- Secured
- Simple Syntax
- High Level Programming Language
- High Performance

JVM- JAVA Virtual Machine



Source: <http://www.programmersnight.com/difference-between-jrejvm-and-jdk/>

JRE-Java runtime environment



Source: <http://www.programmersnight.com/difference-between-jre-vm-and-jdk/>

Structure of java program

Documentation Section
Package Statement
Import Statement
Interface Statement
Class Definition
Main Method Class { //Main method defintion }

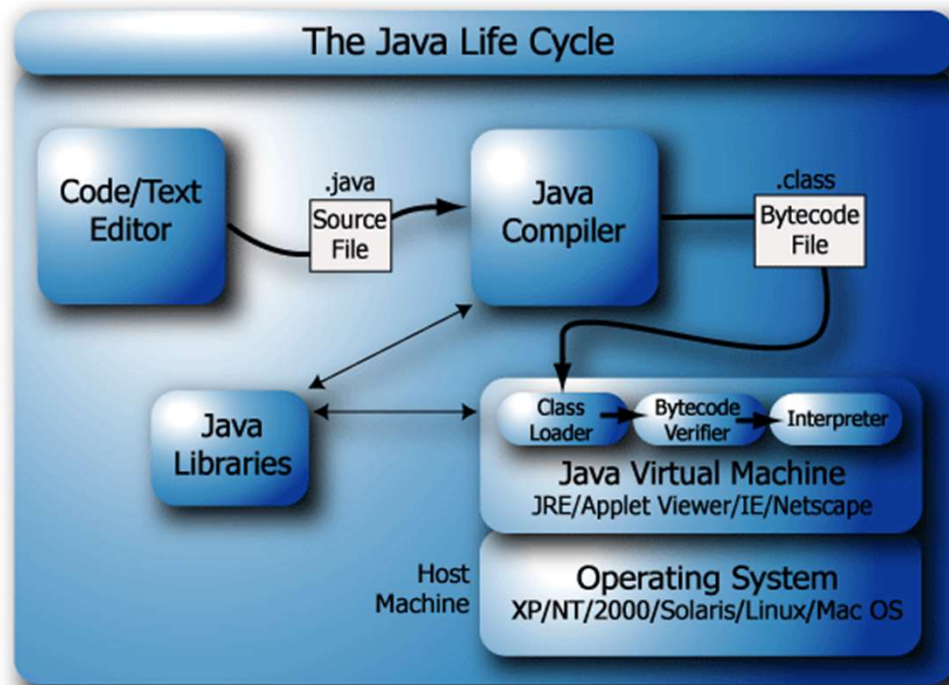
Source: <https://technoleaf.wordpress.com/category/java-basic/java-program-structure/>



Basic JAVA Program

```
public class MyFirstPara {  
    public static void main(String args[]){  
        System.out.println("Hello World!");  
    }  
}
```

JAVA Program lifecycle

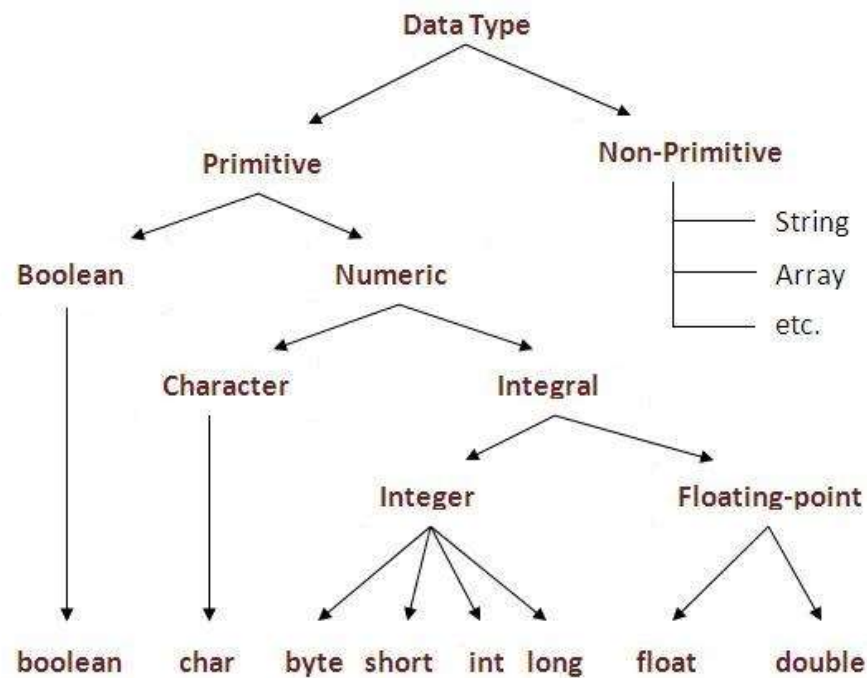


Source: <http://www.eeng.dcu.ie/~ee553/ee402notes/html/ch04s02.html>

Fundamentals of Class

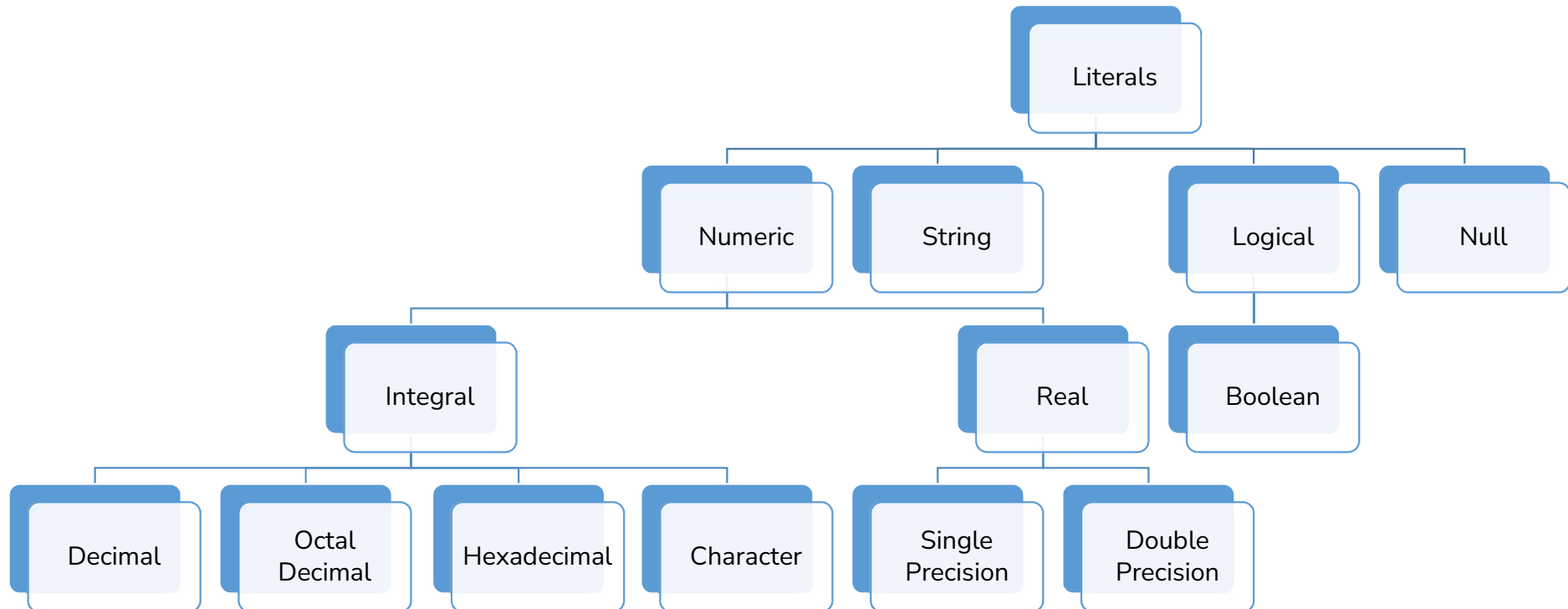
- Difference between Class & Class File.
- Object Creation.
- “new” keyword.
- Constructor Overloading.
- Comments.

Data types in java



Source: <http://www.javatpoint.com/variable-datatype>

Literals in JAVA



```
class Computer {
    Computer() {
        System.out.println("Constructor of Computer class.");
    }

    void computer_method() {
        System.out.println("Power gone! Shut down your PC soon...");
    }

    public static void main(String[] args) {
        Computer my = new Computer();
        Laptop your = new Laptop();

        my.computer_method();
        your.laptop_method();
    }
}

class Laptop {
    Laptop() {
        System.out.println("Constructor of Laptop class.");
    }

    void laptop_method() {
        System.out.println("99% Battery available.");
    }
}
```

Source: <http://www.programmingsimplified.com/java/source-code/java-program-multiple-classes>

Naming conventions

Package names: start with lowercase letter

- E.g. java.util, java.net, java.io ...

Class names: start with uppercase letter

- E.g. File, Math ...
- avoid name conflicts with packages
- avoid name conflicts with standard keywords in java system

Variable, field and method names: start with lowercase letter

- E.g. x, out, abs ...

Constant names: all uppercase letters

- E.g. PI ...

Multi-word names: capitalize the first letter of each word after the first one

- E.g. HelloWorldApp, getName ...

Exception class names: (1) start with uppercase letter (2) end with “Exception” with normal exception and “Error” with fatal exception

- E.g. OutOfMemoryError, FileNotFoundException

Assignment

1. Write an interactive program to compute simple & compound interest.
2. Write an interactive program to find the largest of two numbers.

Hints

- Simple Interest

- Use the following simple interest formula: $I = p \times r \times t$
 - p is the principal or money deposited
 - r is the rate of interest
 - t is time

- Compound Interest

- compound interest is the interest earned not only on the original principal, but also on all interests earned previously.
- In other words, at the end of each year, the interest earned is added to the original amount and the money is reinvested.

Ques?

THANK YOU...